
GENERAL RELATIVITY

Tutorial problem set 13, 13.12.2019.

■ PROBLEM 34 De Sitter-Schwarzschild solution.

This problem is similar to the first problem from last week's problem set. You will solve the Einstein's equation in vacuum ($T_{\mu\nu} = 0$) for the spherically symmetric metric, only now we will assume a non-vanishing cosmological constant $\Lambda > 0$. Start with the following ansatz for the line element,

$$ds^2 = -f(t, r)dt^2 + w(t, r)dr^2 + r^2 d\Omega^2 . \quad (34.1)$$

- (a) Write down all the non-trivial components of the Einstein equation.

The non-vanishing components of the Riemann tensor for line element (34.1) are

$$R^0_{101} = \frac{1}{4f^2w} \left[2fw\partial_t^2 w - f(\partial_t w)^2 - w\partial_t f\partial_t w + f\partial_r f\partial_r w - 2wf\partial_r^2 f + w(\partial_r f)^2 \right] , \quad (34.2)$$

$$R^0_{202} = -\frac{r}{2fw}\partial_r f , \quad R^0_{303} = -\frac{r}{2fw}\sin^2 \vartheta \partial_r f , \quad (34.3)$$

$$R^0_{212} = -\frac{r}{2fw}\partial_t w , \quad R^0_{313} = -\frac{r}{2fw}\sin^2 \vartheta \partial_t w , \quad (34.4)$$

$$R^1_{212} = \frac{r}{2w^2}\partial_r w , \quad R^1_{313} = \frac{r}{2w^2}\sin^2 \vartheta \partial_r w , \quad R^2_{323} = \frac{w-1}{w}\sin^2 \vartheta , \quad (34.5)$$

and the non-vanishing components of the Ricci tensor are R_{00} , R_{11} , R_{22} , R_{33} , and $R_{01} = R_{10}$. You do not need to calculate the Ricci scalar explicitly (from the Riemann tensor), why?

Hint: examine the trace of the Einstein equation.

- (b) Argue from the tr component of the Einstein equation that w is independent of time, $w(t, r) = w(r)$.
- (c) By taking an appropriate linear combination of the remaining non-trivial components of the Einstein equation show that f must be of the form $f(t, r) = F(t)A(r)$, and $A(r) = 1/w(r)$. Further, argue that $F(t)$ can be subsumed into the redefinition of time, and needs not be considered.
- (d) Finally, show that Einstein equation implies

$$A(r) = 1 - \frac{R_S}{r} - \frac{\Lambda r^2}{3} . \quad (34.6)$$

By requiring the standard Schwarzschild limit when $\Lambda \rightarrow 0$ fix the constant R_S .

- (e) In these coordinates the horizons are located at the radii defined by $A(r) = 0$. One can determine their location by solving this condition, which is not so straightforward to solve. It is a cubic equation which can be written in a compact form as

$$x^3 - y^2(x - 1) = 0 , \quad (34.7)$$

where $x = r/R_S$, $y = R_H/R_S$, and $R_H = \sqrt{3/\Lambda}$. There is a regime for y where this condition has no solutions $r > 0$, and a regime for y where it has two solutions $r > 0$. The second regime corresponds to two horizons (a black hole one and a cosmological one). In between these two regimes, there is a critical value $y = y_{cr}$ for which the horizons overlap exactly at $x = x_{cr}$. It means that (34.7) has a double root $r > 0$, and it can be factorized as

$$0 = (x - x_{cr})^2(x - \alpha) , \quad (34.8)$$

where α is some constant smaller than zero, and therefore we are not interested in it. By equating (34.8) with (34.7) for $y = y_{cr}$ it is possible to determine what y_{cr} and x_{cr} are. From these values find the condition that M and Λ have to satisfy in order for the two horizons to exist.

How would you address the question of what the space-time looks like for $y < y_{cr}$?

■ PROBLEM 35 Kruskal-Szekeres coordinates.

In this problem you will show that the singularity at Schwarzschild radius, $R_S = 2MG_N$, of the Schwarzschild metric,

$$ds^2 = - \left(1 - \frac{R_S}{r}\right) dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 , \quad (35.1)$$

can be removed by making use of different coordinates.

We start by defining the *Kruskal-Szekeres* coordinates, for $r > R_S$:

$$R = \left(\frac{r}{R_S} - 1\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \cosh\left(\frac{t}{2R_S}\right) , \quad (35.2)$$

$$T = \left(\frac{r}{R_S} - 1\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \sinh\left(\frac{t}{2R_S}\right) , \quad (35.3)$$

and for $r < R_S$:

$$R = \left(1 - \frac{r}{R_S}\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \sinh\left(\frac{t}{2R_S}\right) , \quad (35.4)$$

$$T = \left(1 - \frac{r}{R_S}\right)^{1/2} \exp\left(\frac{r}{2R_S}\right) \cosh\left(\frac{t}{2R_S}\right) . \quad (35.5)$$

- (a) Show that the metric in Kruskal-Szekeres takes the following form

$$ds^2 = -\frac{4R_S^3}{r} e^{-\frac{r}{R_S}} (dT^2 - dR^2) + r^2 d\Omega^2 . \quad (35.6)$$

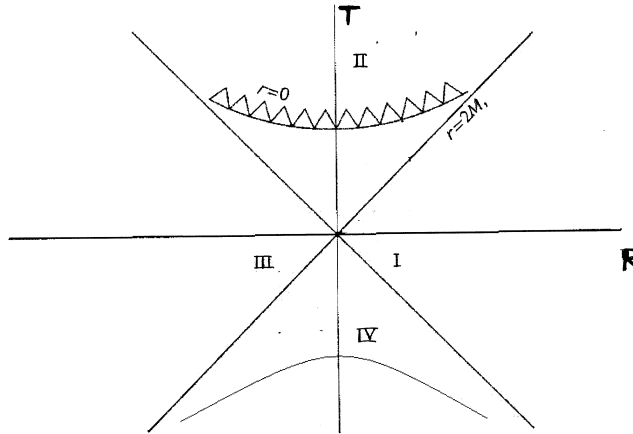


Figure 1: Kruskal diagram.

(b) How is the radial null line parametrized, in terms of R and T ?

In order to get a better understanding of this space-time, we can study the diagram in Fig. 1. Every point in this diagram is supposed to represent a sphere (coordinates ϑ and φ are suppressed). The results from part (b) of this problem tell us that the light cones in coordinates T and R are as open as in SR.

- (c) What is the shape of the lines of constant r , for $r > R_S$? And for $r < R_S$? Can a massive object stay at constant r inside the horizon?
- (d) Draw the lines of constant t . What happens for $t \rightarrow \infty$?
- (e) Draw the path of an infalling observer.
- (f) Which regions are connected to region I by light signals?

Read section 5.7 from Carroll's book for a discussion of the properties of Schwarzschild black hole in Kruskal-Szekeres coordinates.